

2nd semester problem sets

June 10, 2026

1 Problem set 1

Q1. Determine which of the following are subrings of $\mathbb{Q}$

- a. rational numbers with odd denominator
- b. rational numbers with even denominator
- c. nonnegative rationals
- d. squares of rationals
- e. rationals with odd numerator
- f. rationals with even numerator

Soln: Let's check option (a). Let $\mathbb{Q}' = \left\{ \frac{a}{b} \mid a \in \mathbb{Z}, b = 2k+1, \text{ for some integer } k \right\}$

Suppose $\frac{a}{b}$ and $\frac{c}{d}$ are two elements of the set $\mathbb{Q}'$. Then we have that $b = 2k+1$ and $d = 2m+1$ for some integers k, m . Now,

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Since an odd integer times an odd integer is always odd, we get that $\frac{a}{b} - \frac{c}{d} \in \mathbb{Q}'$. Also, clearly, $1 \in \mathbb{Q}'$ and $0 \in \mathbb{Q}'$. So, $\mathbb{Q}'$ is a subring of $\mathbb{Q}$

Q2

Q3: Show the center of a ring is a subring containing identity. Show the center of a division ring is a field.

Soln: Let R be a ring and let $Z(R)$ be the center of R . We want to show that $Z(R)$ is a subring of R containing identity. Let $a, b \in Z(R)$. Clearly, 1 and 0 are in $Z(R)$ since $1 \cdot x = x \cdot 1$ for all $x \in Z(R)$, and similarly for 0. Now let c be an arbitrary element of R . Then, we have $(a - b)c = ac - bc = ca - cb = c(a - b)$. This means that $a - b \in Z(R)$ and thus $Z(R)$ is a subring of R containing identity.

For the second part, we know that a division ring is a ring (not necessarily commutative) such that every element is invertible. If R is a division ring, then $Z(R)$ is a subring of R which is commutative, thus the result follows.

Q4 If R is an integral domain such that $x^2 = 1$, then $x = \pm 1$.

Soln: $x^2 = 1 \implies x^2 - 1 = 0 \implies (x + 1)(x - 1) = 0 \implies x + 1 = 0$ or $x - 1 = 0 \implies x = \pm 1$